

Operating System – Unit-1

1. Operating System

1.1 Definition

- An **Operating System (OS)** is a system software which manages all the hardware resources of a computer and provides an environment in which application programs can run.
- In simple words, the operating system works as an **interface between the user and the hardware**.
- The user gives instructions through programs and the operating system controls the hardware to perform those tasks.

1.2 Why Operating System is needed

An operating system is required because:

- users cannot directly use CPU, memory and devices,
- many programs run at the same time and must be managed properly,
- memory must be shared safely,
- hardware devices must be controlled in an organised way.

Without an operating system, it is almost impossible for a normal user to use a computer system.

1.3 Main functions of Operating System

The important functions of an operating system are:

1. Process management
2. Memory management
3. File system management
4. Device management
5. Security and protection
6. User interface

(1) Process management

The operating system creates, schedules and terminates processes.

It decides which process will get the CPU and for how long.

(2) Memory management

The operating system allocates memory to programs and frees memory after the program finishes.

It also keeps track of used and free memory.

(3) File system management

The operating system manages files and directories.
It controls reading, writing and access permissions of files.

(4) Device management

The operating system controls all I/O devices such as keyboard, disk, printer and display using device drivers.

(5) Protection and security

The operating system protects data and programs and controls access to system resources.

(6) User interface

The operating system provides graphical and command based interfaces.

2. Classification of Operating Systems

2.1 Definition

Classification of operating systems means grouping operating systems on the basis of their working style and features.

2.2 Types of Operating Systems

The main types are:

1. Batch Operating System
2. Interactive Operating System
3. Time Sharing Operating System
4. Real Time Operating System
5. Multiprocessor Operating System
6. Multiuser Operating System
7. Multiprocess Operating System
8. Multithreaded Operating System

3. Batch Operating System

Definition

A **Batch Operating System** is an operating system in which jobs are collected in groups and executed one by one without user interaction.

Detailed explanation

In a batch operating system:

- users submit their jobs to the computer operator,
- jobs are stored in a batch,

- jobs having similar requirements are grouped,
- the operating system executes the jobs sequentially.

Once a job starts, the user cannot communicate with the system until the job finishes.

Main characteristics

- no interaction during execution,
- long waiting time,
- simple job processing method.

Example

Payroll systems and billing systems in early computers.

4. Interactive Operating System

Definition

An **Interactive Operating System** allows the user to interact with the computer while a program is running.

Detailed explanation

In this type of operating system:

- the user enters commands,
- the system responds immediately,
- the user can give new commands while the program is executing.

The main aim of an interactive system is to provide fast and continuous communication between the user and the computer.

Main characteristics

- fast response,
- continuous user interaction,
- suitable for personal computers.

Example

Desktop and laptop systems.

5. Time Sharing Operating System

Definition

A **Time Sharing Operating System** allows many users or many programs to share the CPU by dividing CPU time into small time intervals.

Detailed explanation

In a time sharing system:

- the operating system assigns CPU to a process for a very short time,
- then switches to another process,
- this switching continues for all processes.

Because the switching is very fast, all programs appear to run at the same time. This technique improves system utilisation and user satisfaction.

Important point

This frequent switching of CPU from one process to another is called **context switching**.

Example

Running browser, Python program and editor together on the same computer.

6. Real Time Operating System (RTOS)

Definition

A **Real Time Operating System** is an operating system in which tasks must be completed within a fixed time limit.

Both the result and the time at which the result is produced are important.

Detailed explanation

In a real time operating system:

- tasks are associated with deadlines,
- the operating system schedules tasks in such a way that deadlines are met,
- missing a deadline may lead to serious system failure.

A late result is treated as an incorrect result.

Types of Real Time Operating System

(1) Hard real time system

In a hard real time system, missing a deadline is not acceptable.

(2) Soft real time system

In a soft real time system, small delay can be tolerated.

Examples

- airbag system,
- medical monitoring system,
- industrial robots.

7. Multiprocessor Operating System

Definition

A **Multiprocessor Operating System** is an operating system that supports more than one CPU in a computer system.

Detailed explanation

In this type of system:

- two or more processors are available,
- the operating system distributes processes among the processors,
- tasks can be executed in parallel.

The operating system also handles communication and synchronization between processors.

Advantages

- better performance,
- faster execution,
- higher system throughput.

Example

Server systems with multiple CPUs.

8. Multiuser Operating System

Definition

A **Multiuser Operating System** allows more than one user to use the same computer system at the same time.

Detailed explanation

In a multiuser system:

- each user has a separate session,
- all users share the same hardware resources,
- the operating system ensures fairness and security among users.

Example

University or company server used by many users simultaneously.

9. Multiprocess Operating System

Definition

A **Multiprocess Operating System** allows more than one process to execute at the same time.

Detailed explanation

Each application running on the computer is treated as a separate process. The operating system schedules all these processes on the CPU. It also manages communication and synchronization among processes.

Example

Running browser, music player, Python program and editor together.

10. Multithreaded Operating System

Definition

A **Multithreaded Operating System** supports multiple threads inside a single process.

Detailed explanation

A thread is the smallest unit of execution.

In a multithreaded system:

- all threads of a process share the same memory,
- each thread has its own execution flow,
- threads can run in parallel on multi-core systems.

Multithreading improves responsiveness and resource utilisation.

Engineering example

One thread reads data, another processes data and another saves the output.

11. Operating System Structure

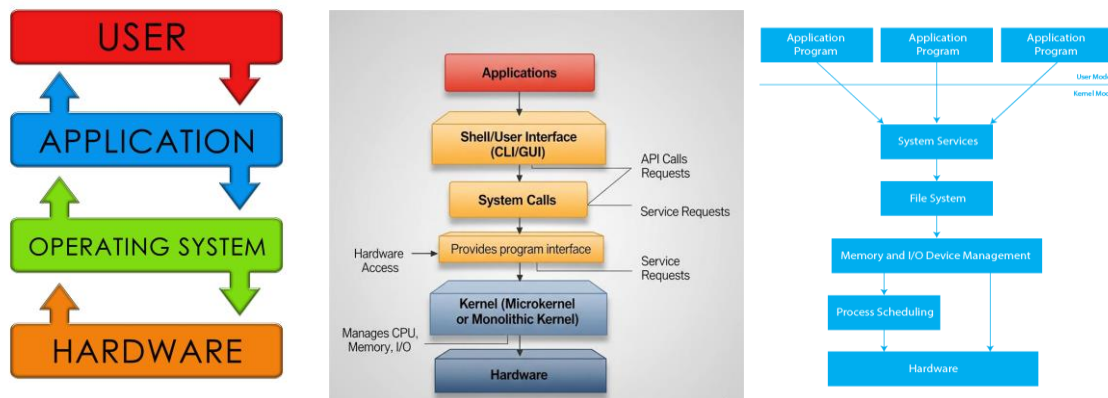
Definition

Operating system structure describes how different components of the operating system are organised.

Types of Operating System Structure

1. Layered structure
2. Monolithic kernel structure
3. Microkernel structure

12. Layered Structure



Definition

In the layered structure, the operating system is divided into several layers. Each layer performs a specific task and uses the services of the lower layer.

Detailed explanation

The lowest layer is hardware.

The highest layer interacts with the user.

A request passes through different layers before reaching the hardware.

Advantages

- easy design,
- easy debugging,
- clear modular structure.

Disadvantages

- additional communication overhead,
- slower performance compared to simple structures.

13. System Components of Operating System

Definition

System components are the main internal parts of the operating system that perform specific functions.

Main components

- process management component,

- memory management component,
- file system component,
- input output component,
- networking component,
- protection and security component.

Each component cooperates with others to provide proper system functionality.

14. Operating System Services

Definition

Operating system services are the services provided by the operating system to users and application programs.

Types of operating system services

(A) User oriented services

- program execution,
- input output operations,
- file manipulation,
- communication between processes,
- error detection.

(B) System oriented services

- resource allocation,
- accounting,
- protection and security.

15. Reentrant Kernel

Definition

A **Reentrant Kernel** is a kernel that can be safely executed by more than one process at the same time.

Detailed explanation

In modern systems, many processes may enter the kernel simultaneously.

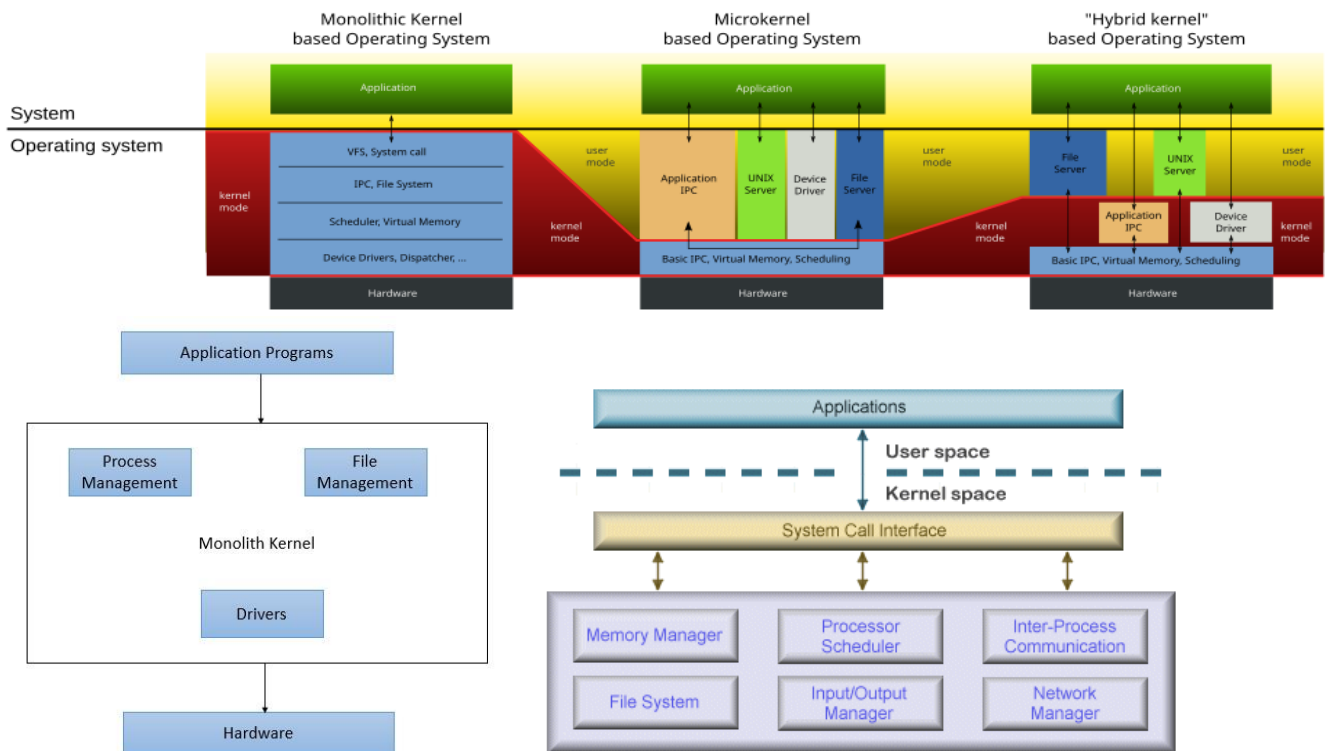
The kernel must ensure that shared data is not corrupted.

A reentrant kernel uses proper synchronization techniques and avoids unsafe shared global data.

Example

Two processes accessing file services at the same time without damaging the file system.

16. Monolithic Kernel System



Definition

In a **Monolithic Kernel System**, all operating system services run inside a single large kernel.

Detailed explanation

Process management, memory management, file system and device drivers all execute in kernel space.

All services communicate directly inside the kernel.

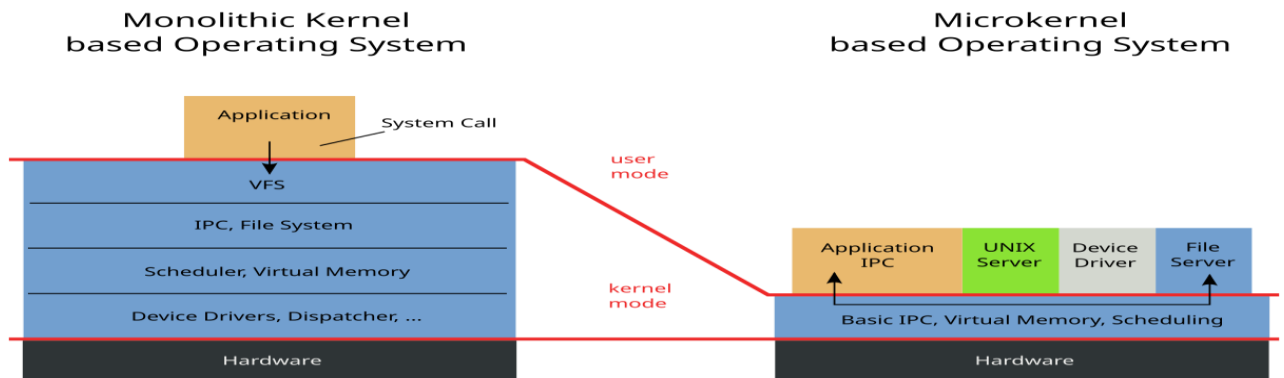
Advantages

- fast execution,
- low overhead,
- high performance.

Disadvantages

- large and complex kernel,
- difficult maintenance,
- one fault can crash the entire system.

17. Microkernel System



Definition

A **Microkernel System** contains only the most essential services inside the kernel and moves other services to user space.

Detailed explanation

The kernel performs only basic functions such as:

- process scheduling,
- memory management,
- inter-process communication.

All other services such as file system and device drivers run as user processes.

Advantages

- better reliability,
- easy debugging,
- better fault isolation.

Disadvantages

- lower performance due to message passing overhead.

✓ 1. Batch OS vs Time Sharing OS

Point	Batch Operating System	Time Sharing Operating System
User interaction	No interaction during execution	Continuous interaction is possible
Execution style	Jobs run one by one in batches	Many processes share CPU
Response time	Very high (slow response)	Very low (fast response)
CPU usage	Often inefficient	Efficient use of CPU

Main use	Large offline jobs	Multi-program / multi-user systems
Example	Payroll, billing jobs	Laptop / desktop system

✓ 2. Multiprocessor OS vs Multiprocess OS

Point	Multiprocessor OS	Multiprocess OS
Meaning	System has more than one CPU	System runs more than one process
Focus	Hardware based	Software / execution based
CPU count	Two or more CPUs	Can run on single CPU also
Main idea	Parallel processing using multiple CPUs	Multiple programs executing
Example	Server with 2 processors	Browser + Python + editor

👉 **Short line to remember**

✓ Multiprocessor → multiple CPUs

✓ Multiprocess → multiple processes

✓ 3. Multiuser OS vs Multitasking (Multiprocess) OS

Point	Multiuser OS	Multitasking / Multiprocess OS
Main focus	Multiple users	Multiple processes
Who is multiple?	Users	Programs / processes
Sessions	Each user has a session	Same user can run many programs
Typical use	Server systems	Personal computers
Example	College server	Laptop running many apps

✓ 4. Process vs Thread

Point	Process	Thread
Definition	Program in execution	Smallest unit of execution inside a process
Memory	Has its own memory	Shares memory with other threads
Creation cost	High	Low
Communication	Slow (IPC needed)	Fast (shared memory)
Failure effect	One process crash does not affect others	One bad thread can affect the process
Example	Python program	Data loader thread, training thread

✓ 5. Monolithic Kernel vs Microkernel

Point	Monolithic Kernel	Microkernel
Kernel size	Large	Small
Where services run	Inside kernel	Mostly in user space
Performance	High	Slightly lower
Reliability	Lower	Higher
Maintenance	Difficult	Easier
Communication	Direct function calls	Message passing
Fault isolation	Poor	Good

✓ 6. Hard Real-Time OS vs Soft Real-Time OS

Point	Hard RTOS	Soft RTOS
Deadline	Must be met	Can be missed sometimes
Failure impact	System failure	Quality degrades only
Criticality	Very high	Medium
Usage	Safety-critical systems	Multimedia / streaming
Example	Airbag system	Video / audio system

7. Interactive OS vs Batch OS

Point	Interactive OS	Batch OS
User interaction	Allowed	Not allowed
Response	Immediate	After job completion
Execution	One command at a time with feedback	Job oriented
Waiting time	Low	High
Example	Desktop system	Old mainframe systems

8. Time Sharing OS vs Real Time OS

Point	Time Sharing OS	Real Time OS
Goal	Fair CPU sharing	Guaranteed response time
Deadline	Not strict	Strict
Late response	Acceptable	Not acceptable
Use case	General purpose computing	Control systems
Example	Laptop OS	Robot / medical system